PRE-INTERVENTION BASELINE KNOWLEDGE TEST

Knowledge of Social Media Algorithms

Participant Information

Participant Code: _____

Date: _____

Purpose of the Assessment

This assessment is designed to measure your current knowledge of social media algorithms before participating in the professional development intervention. Your responses will be used for research purposes only and will be treated confidentially.

This is not an examination of your teaching performance. Please answer each question based on your current knowledge.

Instructions

1. The assessment contains 16 questions divided into four sections.
2. For multiple-choice questions, select only one answer by ticking the appropriate box.
3. For True/False questions, tick either True or False.
4. Answer all questions without consulting books, websites, notes, or other participants.

Suggested completion time: 30 minutes

Section A: Content Knowledge

CK1. Social media algorithms determine which content to display primarily based on:

Response option	Tick one
A. Random selection of all available content	☐
B. Users' past interactions, content types, and engagement patterns	☐
C. Only the user's friend network	☐
D. The chronological order of posts	☐

CK2. A user frequently likes and shares cooking videos. How will the algorithm most likely respond?

Response option	Tick one
A. It will show fewer cooking videos to avoid repetition	☐
B. It will increase recommendations of cooking and related content	☐
C. It will show only trending videos unrelated to cooking	☐
D. It will ignore the user's behaviour	☐

CK3. Social media algorithms always give users the content they explicitly requested.

True	False
☐	☐

CK4. Users' interactions, such as clicks, likes, comments, and shares, influence the content that an algorithm predicts they will enjoy.

True	False
☐	☐

Section B: Pedagogical Content Knowledge

PCK1. When teaching students about algorithmic recommendations, which approach is most effective?

Response option	Tick one
A. Lecturing on definitions and terminology only	☐
B. Providing relatable scenarios, simulations, or guided examples	☐
C. Assuming that students will understand algorithms intuitively	☐
D. Asking students to memorise a list of algorithm types	☐

PCK2. To explain social media algorithms inclusively, a teacher should:

Response option	Tick one
A. Use only examples that apply to the majority of students	☐
B. Use multiple teaching methods and adapt examples to different learners	☐
C. Focus mainly on high-achieving students	☐
D. Avoid using students' experiences	☐

PCK3. Teachers can improve students' understanding of algorithms by connecting lessons to students' everyday social media experiences.

True	False
☐	☐

PCK4. Inclusive teaching only involves simplifying content for students who experience learning difficulties.

True	False
☐	☐

Section C: Technological Content Knowledge

TCK1. Which statement best describes how social media algorithms use data?

Response option	Tick one
A. They consider all types of user data equally	☐
B. They prioritise user interactions, inferred interests, and metadata when recommending content	☐
C. They consider only information that users explicitly post	☐
D. They operate independently of users' activities	☐

TCK2. Which of the following is an example of inferred data?

Response option	Tick one
A. A user's stated favourite colour	☐
B. Interests predicted from the user's browsing and engagement patterns	☐
C. The number of posts shared by the user	☐
D. The user's email address	☐

TCK3. Social media algorithms are technological systems that influence the visibility and ordering of content on social media platforms.

True	False
☐	☐

TCK4. Social media platforms never infer additional information about users beyond the information that users provide directly.

True	False
☐	☐

Section D: Technological Pedagogical Knowledge

TPK1. To teach social media algorithms effectively using technology, the best approach is to:

Response option	Tick one
A. Create a simulated content feed that demonstrates algorithmic behaviour	☐
B. Lecture about algorithms without providing examples	☐
C. Ask students to memorise definitions from a textbook	☐
D. Show random videos that are unrelated to the lesson	☐

TPK2. Which activity would best help students understand algorithmic personalisation?

Response option	Tick one
A. Asking students to map their social media feeds and predict what the algorithm may show next	☐
B. Asking students to write definitions only	☐
C. Asking students to watch unrelated videos	☐
D. Asking students to listen to a lecture without participating in an activity	☐

TPK3. Integrating content, pedagogy, and technology can help students develop a deeper understanding of social media algorithms.

True	**False**
☐	☐

TPK4. The integration of technology does not significantly affect students' learning about social media algorithms.

True	**False**
☐	☐

POST-INTERVENTION KNOWLEDGE TEST

Knowledge of Social Media Algorithms

Participant Information

Participant Code: _____

Date: _____

Purpose of the Assessment

This assessment is designed to measure your knowledge of social media algorithms after participating in the professional development intervention.

Your responses will be used for research purposes only and will be treated confidentially. The assessment is not intended to evaluate your general teaching performance.

Instructions

1. This assessment contains 16 questions divided into four sections.
2. For multiple-choice questions, select only one answer by ticking the appropriate box.
3. For True/False questions, tick either True or False.
4. Answer all questions based on the knowledge and skills you gained during the intervention.
5. Do not consult books, websites, notes, training materials, or other participants while completing the assessment.

Suggested completion time: 30 minutes

Section A: Content Knowledge

CK1. User actions such as clicks, likes, comments, and shares do not affect the content recommended by social media algorithms.

True	False
☐	☐

CK2. If a user frequently likes and shares cooking-related videos, how is the algorithm most likely to respond?

Response option	Tick one
A. It ignores the user's behaviour	☐
B. It shows mostly trending content unrelated to cooking	☐
C. It increases the amount of cooking and similar content recommended	☐
D. It reduces cooking-related content to avoid repetition	☐

CK3. What is the primary basis on which social media algorithms select content to display?

Response option	Tick one
A. The order in which posts are published	☐
B. Users' previous interactions, content types, and engagement patterns	☐
C. Random selection from all available content	☐
D. Only the user's network of friends	☐

CK4. Social media algorithms do not always provide users with only the content they explicitly request.

True	False
☐	☐

Section B: Pedagogical Content Knowledge

PCK1. Inclusive teaching involves more than simply simplifying content for students who require additional learning support.

True	False
☐	☐

PCK2. Which teaching approach is most effective for helping students understand algorithmic recommendations?

Response option	Tick one
A. Requiring students to memorise different types of algorithms	☐
B. Using relatable examples, simulations, or guided learning activities	☐
C. Assuming students will understand the concept independently	☐
D. Focusing only on definitions and theoretical explanations	☐

PCK3. To teach social media algorithms inclusively, a teacher should:

Response option	Tick one
A. Focus mainly on high-achieving students	☐
B. Avoid incorporating students' experiences into the lesson	☐
C. Use a variety of teaching strategies and adapt examples to suit different learners	☐
D. Use examples that apply only to the majority of students	☐

PCK4. Connecting algorithm concepts to students' everyday social media experiences can improve their understanding.

True False

☐ ☐

Section C: Technological Content Knowledge

TCK1. Social media platforms can infer additional information about users beyond the information that users explicitly provide.

True False

☐ ☐

TCK2. Which of the following best represents inferred data?

Response option	Tick one
A. The number of posts shared by a user	☐
B. A user's email address	☐
C. Interests predicted from the user's browsing and interaction patterns	☐
D. A user's stated favourite colour	☐

TCK3. How do social media algorithms typically use data to recommend content?

Response option	Tick one
A. They function independently of user activity	☐
B. They rely only on content explicitly posted by the user	☐
C. They prioritise user interactions, inferred interests, and metadata	☐
D. They treat all types of user data equally without prioritisation	☐

TCK4. Social media algorithms influence which content is made visible to users.

True False

☐ ☐

Section D: Technological Pedagogical Knowledge

TPK1. Integrating technology into instruction has little effect on students' understanding of social media algorithms.

True False

☐ ☐

TPK2. Which activity is most effective in helping students understand how algorithmic personalisation works?

Response option	Tick one
A. Listening to a lecture without interaction	☐
B. Watching videos unrelated to the lesson	☐
C. Analysing content feeds and predicting how the algorithm may behave	☐
D. Writing definitions without applying them	☐

TPK3. What is the most effective way to teach social media algorithms using technology?

Response option	Tick one
A. Showing digital content that is unrelated to the lesson	☐
B. Asking students to memorise textbook definitions	☐
C. Using simulations that demonstrate how social media algorithms work	☐
D. Relying only on verbal explanations	☐

TPK4. Combining content knowledge, pedagogy, and technology enhances students' understanding of social media algorithms.

True	False
☐	☐

PERCEIVED KNOWLEDGE OF TEACHING SOCIAL MEDIA ALGORITHMS

Participant Information

Participant Code: _____

Date: _____

Purpose of the Questionnaire

This questionnaire is designed to assess changes in your perceived knowledge of social media algorithms and their teaching after completing the professional development programme.

You are asked to reflect on your current understanding and compare it with the level of knowledge you believe you had before participating in the programme.

Your responses will be used for research purposes only and will be treated confidentially. There are no right or wrong answers. Please provide responses that honestly reflect your own knowledge and experience.

Instructions

For each statement, provide two ratings:

1. Before Intervention: Rate the level of knowledge you believe you had before participating in the professional development programme.
2. After Intervention: Rate your current level of knowledge after completing the professional development programme.

Please tick only one response in each column for every statement.

Response Scale

Score	Level of knowledge
1	Not at all knowledgeable
2	Slightly knowledgeable
3	Moderately knowledgeable
4	Very knowledgeable
5	Extremely knowledgeable

Section A: Content Knowledge

This section assesses your understanding of social media algorithms and how they function.

Item	Statement	Before Intervention	After Intervention
CK1	I understand what social media algorithms are and their primary purpose.	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐
CK2	I understand how social media platforms personalise content for users.	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

CK3	I understand how users' interactions, such as likes, views, comments, and shares, influence algorithmic recommendations.	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐
CK4	I understand how social media algorithms can shape users' exposure to information, including the creation of filter bubbles or echo chambers.	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

Section B: Pedagogical Content Knowledge

This section assesses your knowledge of effective and inclusive strategies for teaching social media algorithms.

Item	Statement	Before Intervention	After Intervention
PCK1	I understand effective teaching strategies for teaching social media algorithms.	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐
PCK2	I can explain social media algorithm concepts to students in an age-appropriate manner.	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐
PCK3	I can teach social media algorithm concepts inclusively in ways that accommodate students with diverse learning needs.	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐
PCK4	I can design learning activities, such as simulations, case studies, or real-life scenarios, that help students understand how social media algorithms operate.	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

Section C: Technological Content Knowledge

This section assesses your understanding of the technological systems and processes behind social media algorithms.

Item	Statement	Before Intervention	After Intervention
TCK1	I understand how algorithmic technologies influence the visibility and ordering of content on social media platforms.	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐
TCK2	I understand how data collection and algorithmic systems interact within social media platforms.	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐
TCK3	I can identify examples of algorithmic curation and recommendation systems used by social media platforms.	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐
TCK4	I understand how algorithmic techniques, such as ranking, filtering, and predictive modelling, are used to organise and prioritise social media content.	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

Section D: Technological Pedagogical Knowledge

This section assesses your ability to integrate technology into the teaching of social media algorithms.

Item	Statement	Before Intervention	After Intervention
TPK1	I can use digital tools, simulations, or social media platform examples to support teaching about social media algorithms.	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

TPK2	I can design technology-supported learning activities that help students understand social media algorithms.	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐
TPK3	I feel prepared to integrate social media algorithm concepts into my teaching using appropriate technologies.	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐
TPK4	I can manage and guide students' use of digital tools or platforms to support meaningful, safe, and responsible learning about social media algorithms.	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐	1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐